PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints. (Note- Create Table, insert data as per your Choice)

```
mysql> create table employee
    -> (emp_id int primary key,
    -> emp_name varchar(30),
    -> dept_id int,
    -> foreign key (dept_id) references department (dept_id));
Query OK, 0 rows affected (0.05 sec)

mysql> describe employee;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int         | NO   | PRI | NULL    |       |
| emp_name | varchar(30) | YES  |     | NULL    |       |
| dept_id  | int         | YES  | MUL | NULL    |       |
+----------+-------------+------+-----+---------+-------+
```

```
mysql> create table department
    -> (dept_id int primary key,
    -> dept_name varchar(30) not null);
Query OK, 0 rows affected (0.05 sec)

mysql> describe department;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| dept_id   | int         | NO   | PRI | NULL    |       |
| dept_name | varchar(30) | NO   |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
```

2. Write a SQL query to add any new column to an existing table.

```
mysql> alter table employee
    -> add column salary int not null;
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> describe employee;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int         | NO   | PRI | NULL    |       |
| emp_name | varchar(30) | YES  |     | NULL    |       |
| dept_id  | int         | YES  | MUL | NULL    |       |
| salary   | int         | NO   |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

```
mysql> insert into employee
    -> (emp_id, emp_name, dept_id, salary)
    -> values
    -> (201, 'Alice', 101, 55000),
    -> (202, 'Bob', 103, 40000),
    -> (203, 'Casey', 104, 35000),
    -> (204, 'Donald', 103, 60000),
    -> (205, 'Emanual', 101, 65000);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  55000 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     103 |  60000 |
|    205 | Emanual  |     101 |  65000 |
+--------+----------+---------+--------+
5 rows in set (0.00 sec)
```

```
mysql> insert into department
    -> (dept_id, dept_name)
    -> values
    -> (101, 'IT'),
    -> (102, 'HR'),
    -> (103, 'Testing'),
    -> (104, 'Customer Support');
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+------------------+
| dept_id | dept_name        |
+---------+------------------+
|     101 | IT               |
|     102 | HR               |
|     103 | Testing          |
|     104 | Customer Support |
+---------+------------------+
4 rows in set (0.00 sec)
```

4. Write a SQL query to delete all records from a table.

```
mysql> delete from employee;
Query OK, 6 rows affected (0.04 sec)

mysql> delete from department;
Query OK, 5 rows affected (0.03 sec)
```

5. Write a SQL query to update salary records from a table where id = 205.

```
mysql> update employee
    -> set salary = 70000
    -> where emp_id = 205;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  55000 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     103 |  60000 |
|    205 | Emanual  |     101 |  70000 |
+--------+----------+---------+--------+
5 rows in set (0.00 sec)
```

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

```
mysql> grant select on employee to readonly_user;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> create user 'readonly_user'@'localhost'
    -> identified by 'Readonly@123';
Query OK, 0 rows affected (0.06 sec)

mysql> grant select on company.employee
    -> to 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.03 sec)

mysql> show grant for 'readonly_user'@'localhost';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual
 for the right syntax to use near 'grant for 'readonly_user'@'localhost'' a
mysql> show grants for 'readonly_user'@'localhost';
+-----------------------------------------------------------------------+
| Grants for readonly_user@localhost                                    |
+-----------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `readonly_user`@`localhost`                     |
| GRANT SELECT ON `company`.`employee` TO `readonly_user`@`localhost` |
+-----------------------------------------------------------------------+
2 rows in set (0.00 sec)
```

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

```
mysql> create user 'temp_user'@'localhost'
    -> identified by 'Temp@123';
Query OK, 0 rows affected (0.08 sec)

mysql> revoke select on company.employee
    -> from 'temp_user'@'localhost';
ERROR 1147 (42000): There is no such grant de
mysql> grant insert on company.employee
    -> to 'temp_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> revoke insert on company.employee
    -> from 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

8. Write a query to insert a new employee into the Emp table and commit the transaction.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into employee
    -> (emp_id, emp_name, dept_id, salary)
    -> values
    -> (206, 'Franses', 102, 55000);
Query OK, 1 row affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  55000 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     103 |  60000 |
|    205 | Emanual  |     101 |  70000 |
|    206 | Franses  |     102 |  55000 |
+--------+----------+---------+--------+
6 rows in set (0.00 sec)
```

9. Write a query to update the Salary of an employee with EmpID = 201 and commit the transaction.

```
mysql> update employee
    -> set salary = 65000
    -> where emp_id = 201;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  65000 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     103 |  60000 |
|    205 | Emanual  |     101 |  70000 |
|    206 | Franses  |     102 |  55000 |
+--------+----------+---------+--------+
6 rows in set (0.00 sec)
```

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

11. Write a query to rollback to the previously created SAVEPOINT.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint before_update;
Query OK, 0 rows affected (0.02 sec)

mysql> update employee
    -> set dept_id = 102
    ->  where emp_id = 204;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> rollback to before_update;
Query OK, 0 rows affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  65000 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     102 |  60000 |
|    205 | Emanual  |     101 |  70000 |
|    206 | Franses  |     102 |  55000 |
+--------+----------+---------+--------+
6 rows in set (0.00 sec)
```

12. Write a query to truncate all data from the Emp table. **

```
mysql> truncate table employee;
Query OK, 0 rows affected (0.09 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.

```
mysql> select * from employee
    -> where salary > 50000;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  65000 |
|    204 | Donald   |     102 |  60000 |
|    205 | Emanual  |     101 |  70000 |
|    206 | Franses  |     102 |  55000 |
+--------+----------+---------+--------+
4 rows in set (0.02 sec)
```

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.

```
mysql> update employee
    -> set salary = salary * 1.10
    -> where dept_id = 101;
Query OK, 2 rows affected (0.03 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> select * from employee;
+--------+----------+---------+--------+
| emp_id | emp_name | dept_id | salary |
+--------+----------+---------+--------+
|    201 | Alice    |     101 |  71500 |
|    202 | Bob      |     103 |  40000 |
|    203 | Casey    |     104 |  35000 |
|    204 | Donald   |     102 |  60000 |
|    205 | Emanual  |     101 |  77000 |
|    206 | Franses  |     102 |  55000 |
+--------+----------+---------+--------+
6 rows in set (0.00 sec)
```

15. Write a query to count the total number of employees in each department.

16. Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai. **

```
mysql> CREATE VIEW view1 AS
    -> SELECT e.*
    -> FROM employee e
    -> JOIN department d
    -> ON e.dept_id = d.dept_id
    -> WHERE e.city = 'Mumbai'
    -> AND d.department_name = 'Salesperson';
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM view1;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
1 row in set (0.01 sec)
```

17. Write a query to display the highest salary among all employees.

```
mysql> select max(salary) from employee;
+--------------+
| max(salary) |
+--------------+
|       77000 |
+--------------+
1 row in set (0.01 sec)
```

18. Write a query to display the 2nd highest salary among all employees.

```
mysql> select max(salary) as second_highest
    -> from employee where salary < (
    -> select max(salary) from employee);
+----------------+
| second_highest |
+----------------+
|          71500 |
+----------------+
1 row in set (0.02 sec)
```

19. Write a query to show all employees ordered by JoiningDate in descending order.

```
mysql> select * from employee
    -> order by joining_date desc;
+--------+----------+---------+--------+--------------+
| emp_id | emp_name | dept_id | salary | joining_date |
+--------+----------+---------+--------+--------------+
|    206 | Franses  |     102 |  55000 | 2025-10-09   |
|    202 | Bob      |     103 |  40000 | 2024-05-01   |
|    201 | Alice    |     101 |  71500 | 2024-04-03   |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   |
|    203 | Casey    |     104 |  35000 | 2022-11-10   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   |
+--------+----------+---------+--------+--------------+
6 rows in set (0.00 sec)
```

20. Write a query to rename the DeptName column to DepartmentName.

```
mysql> alter table department
    ->  change column dept_name department_name varchar(50);
Query OK, 0 rows affected (0.18 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+-----------------+
| dept_id | department_name  |
+---------+-----------------+
|     101 | IT              |
|     102 | HR              |
|     103 | Testing         |
|     104 | Customer Support |
+---------+-----------------+
4 rows in set (0.00 sec)
```